SocialCoach: Personalized Social Skill Learning with RL-based Agentic Tutoring and Practice

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Abstract

Social skills such as negotiation and leadership are crucial for personal and professional success in today's interconnected world. However, scalable and effective training remains a significant challenge due to the scarcity of expert coaching. In this paper, we introduce SocialCoach, a holistic LLM-powered agentic tutoring system for personalized social skill development at scale. First, SocialCoach automatically constructs a pedagogically-grounded, theory-to-practice knowledge corpus from diverse expert sources, leveraging a multi-agent pipeline. Second, to personalize the learning journey, it employs an adaptive practice scheduling module that follows a prescription-retrieval-adaptation process. To maximize the long-term learning experience while overcoming the cold-start problem, this policy is optimized within a learner simulation environment through reinforcement learning. Finally, SocialCoach integrates immersive, goal-driven practice, causality-driven proficiency assessment and knowledge-grounded, reflective tutoring to help address the knowing-doing gap. We deploy it in our product, EQoach, and conduct extensive experiments. The results show that SocialCoach improves simulated pathway quality and judge-rated tutoring quality over baseline approaches, while early user feedback indicates strong perceived engagement and usefulness. These findings suggest a practical architecture for personalized and gamified pedagogical platforms on soft skill learning.

CCS Concepts

• **Information systems → Data mining**; • **Social and professional topics → Computing education**.

Keywords

Intelligent Education, Social and Emotional Learning, LLM Agent.

1 Introduction

In this increasingly interconnected and collaborative global landscape, proficiency in complex social skills is critical for both personal well-being and professional success. As illustrated in Figure 1(a), social skills, such as negotiation, leadership, and conflict resolution, profoundly shape an individual's ability to navigate interpersonal environments [7]. These competencies not only underpin effective interpersonal relationships but also serve as significant drivers of economic value. Yet, despite their undeniable importance, the effective acquisition of these skills remains a critical gap for most learners worldwide. Traditionally, expert coaching is a proven pathway for social skill development. As illustrated in Figure 1(b), this educational loop consists of several key stages, including the knowledge preparation for specialized skills, tailored curriculum design, scenario simulation for practice and providing constructive guidance. However, effectively orchestrating these stages requires substantial human expertise and resources, which limits the scalability and accessibility of high-quality coaching [29].

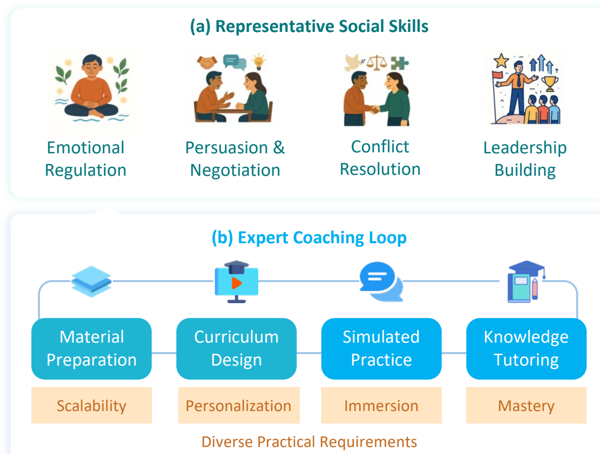

Figure 1: The social skills training landscape. (a) Examples of critical social skills. (b) The typical expert coaching loop, highlighting diverse practical requirements.

The advent of Artificial Intelligence (AI) presents a transformative opportunity to address these limitations. However, early AI-driven methods often selectively focus on isolated educational tasks such as social dialogue generation or knowledge tutoring, and have limited generalizability over various topics. Large Language

Models (LLMs) have demonstrated a remarkable capacity for both vast knowledge integration and context-aware dialogue generation. This is crucial to distinguish their application in declarative knowledge acquisition from the experiential, practice-driven nature of social skill development [44]. Several works have begun to leverage LLM for social conversational practice [17, 22] and to develop frameworks for specific competencies like conflict resolution [35] and empathy [38]. Despite these advances, existing approaches typically target individual coaching stages or specific social skills and do not fully meet the practical requirements. Consequently, a significant gap persists in the absence of a unified social skill development framework that can support comprehensive social skills, adaptive scenario practice, and pedagogically grounded tutoring. This highlights a clear need for a holistic framework that enables scalable, personalized training experiences.

To build such a framework, three fundamental data science challenges must be addressed [21]: (a) *Scalable Knowledge Discovery and Structuring.* To ensure pedagogical rigor, a trustworthy knowledge base must be grounded in verifiable, expert-endorsed principles. However, while LLM-generated content risks unverifiability and hallucination, expert knowledge in social coaching is predominantly unstructured and tacit, often residing in practitioner experience or scattered sources. Thus, a core challenge lies in automatically extracting, codifying, and structuring validated expertise from authoritative sources at scale. (b) *Personalized Practice Curricula Scheduling.* Real-world learners exhibit substantial heterogeneity in their backgrounds, target skills, proficiency levels, and learning paces. Static, one-size-fits-all curricula are thus pedagogically inefficient and hinder learner engagement. Therefore, it is critical but challenging to design an adaptive system that personalizes practice curricula, dynamically tailoring the learning trajectory to each individual's needs and progress. (c) *Diagnostic Coaching Scaffolding Development.* Effective tutoring aims to promote learner autonomy in real-world social skill application. This requires accurately assessing the knowledge deficits in practices and scaffolding to bridge the "knowing–doing" gap. Besides, beyond prescriptive feedback alone, the system must enable guided reflection by linking practice to theory and diagnosing skill gaps. Delivering such diagnostic, reflective guidance remains an underexplored challenge.

To bridge these critical gaps, we introduce **SocialCoach**, a novel agentic tutoring and practice system for personalized social skill training. To ensure educational validity at scale, we develop a structured framework for organizing social skill knowledge, grounded in a theory-to-practice paradigm and enriched through multi-faceted semantic associations. This is implemented via an automated pipeline that transforms unstructured expert input, such as books and papers, into a coherent, structured corpus of strategic theories, illustrative cases, and practical scenarios. This allows the system to support both scalable knowledge retrieval and downstream personalization tasks are central to agentic tutoring. Second, to deliver a personalized learning experience, we model the practice scheduling task as a self-rewarding reinforcement learning (RL) problem. To overcome the classic "cold-start" challenge in personalization, we introduce a high-fidelity LLM-based user simulator that generates realistic interaction data and provides pedagogically relevant reward signals. This allows our system to efficiently learn a policy that dynamically creates and sequences practice scenarios tailored to each user's evolving proficiency profile. Finally, to foster deep and transferable learning, we introduce a pedagogically grounded coaching loop that tightly integrates interactive practice with metacognitive scaffolding. After a user engages in a goal-driven simulated scenario, the system performs a fine-grained diagnostic assessment of their proficiency and provides tailored guidance.

Through a series of component-wise evaluations, case studies, and early product feedback, we show that SocialCoach improves simulated pathway quality and judge-rated tutoring quality over baseline approaches, while users report positive engagement and perceived usefulness. Overall, SocialCoach offers a practical architecture for agentic educational tools in soft-skill domains by integrating key layers of intelligent pedagogy.

We highlight our main contributions as follows.

- We propose a holistic agentic framework for personalized social skill training, SocialCoach. Grounded by pedagogical scaffolding, it closes the coaching loop of interactive practice, proficiency assessment, and tailored tutoring.
- We systematically establish a theory-to-practice social knowledge framework featuring multi-faceted semantic categorization, and automatically construct a traceable knowledge corpus for social skill training via an LLM-powered pipeline.
- To enhance the personalized and immersive experience, we formulate the practice curriculum scheduling as a conditional generative task. We also optimize it via a simulation-based RL to effectively overcome the cold-start challenge.
- We deployed SocialCoach in our practical product and conducted automated evaluations, case studies, and an early user study. Results suggest promising knowledge quality, simulated pathway quality, judge-rated tutoring quality, and perceived user engagement.

2 Related Work

LLM-powered Tutoring Systems. LLMs are increasingly being leveraged to enhance Intelligent Tutoring Systems (ITS), which serve as conversational partners, moving beyond the static, content-centric traditional platforms [5, 20]. Prior work has explored specialized functions within ITS, including learner profiling [46] and performance assessment [33]. Also, there are some studies on enhancing pedagogical strategies of chatbot tutors like Socratic guidance [9, 25]. Furthermore, several multi-agent architectures [30, 40] have been proposed to orchestrate complementary pedagogical roles, where specialized agents collaborate on tasks such as content creation, feedback, and evaluation. Despite these advances, most LLM-powered tutors remain oriented toward knowledge transmission rather than practice-based development. However, social skill learning emphasizes iterative practice, contextual sensitivity, and behavioral transfer. Thus, a critical gap remains in developing scalable, personalized, and pedagogically grounded LLM-driven tutoring and practice systems tailored to social skill learning [44].

Intelligent Social Emotional Learning. Social–emotional learning (SEL) focuses on helping learners cultivate positive attitudes, self-awareness, and healthy interpersonal relationships [10]. Early AI-driven efforts in this space emphasized conversational support, real-time feedback, and scaffolds for socio-emotional growth [34]. With the rise of LLMs, the field has shifted toward adaptive, role-playing–based systems that can target specific skills with greater

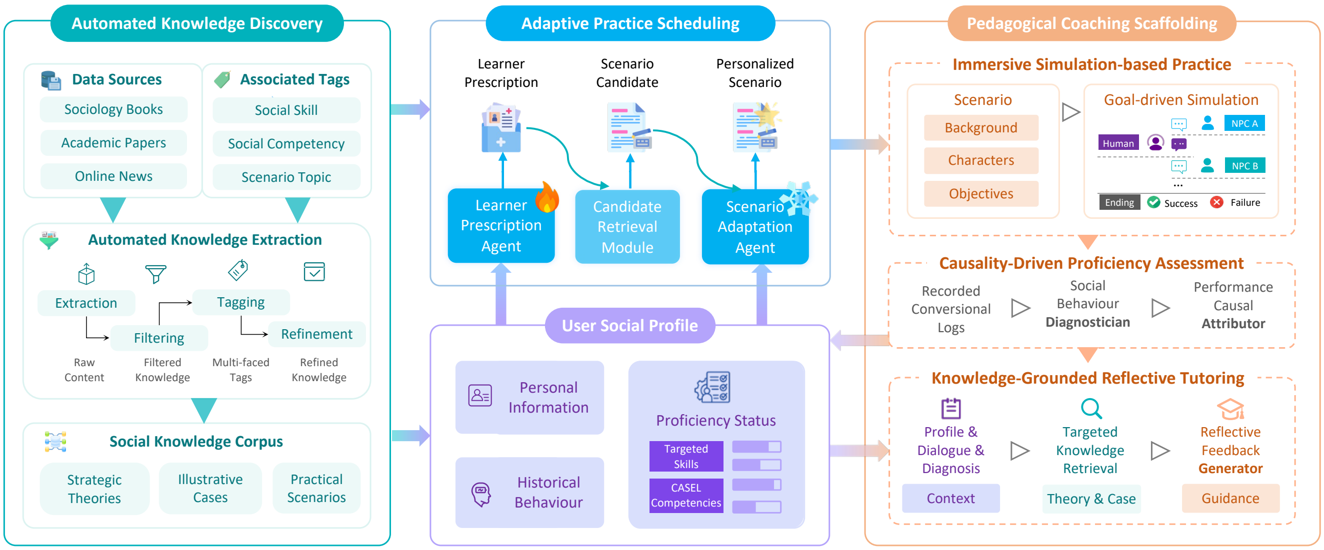

Figure 2: An overview of the SocialCoach Framework for personalized social skill training at scale.

nuance [44]. Recent LLM-based approaches have demonstrated promise in skills such as conflict resolution [35], empathy coaching [38], and stress self-regulation [12] However, despite these advances, current systems are typically scoped to narrow skill areas and face persistent challenges around knowledge traceability and pedagogical reliability. This fragmentation limits the scalability and pedagogical grounding of learning experiences of social skill learning. It also hinders cross-context skill transfer, creating a "knowing–doing" gap between understanding and practical use.

Personalization in Educational System. Personalized learning experience is vital for improving efficiency and engagement as learners progress [27]. Particularly, designing adaptive learning pathways can better match individual needs than traditional or rule-based curricula. To improve path personalization, prior work [6, 15] trains models for specific scenarios using supervised learning on well-labeled data. However, such data are often proprietary or scarce in many educational scenarios. To address such data scarcity and cold-start problem, recent studies [2, 24] use learner simulators as an environment and apply reinforcement learning (RL) to optimize learning path scheduling policies. However, these simulators are typically rule-based and designed for well-structured domains like mathematics and computer science [2]. The personalization of social skill development is further challenging, characterized by diverse learner personas and complex situational contexts. Therefore, optimizing personalized learning pathways in this domain remains a gap, such as building highly representative simulators.

3 Problem Definition

Let $\mathcal{S}$ denote the set of available social skills, and $\mathcal{K}$ represent a verifiable social knowledge corpus. A user is characterized by an initial social profile U^0, which includes the persona information, target skills $S_U \subseteq \mathcal{S}$, and the initial social proficiency level U_p^0. At each learning session t, based on the current user's social profile U^t, a practice scheduling policy π schedules the most appropriate scenario p^t based on the knowledge corpus $\mathcal{K}$. Following the interaction with the scenario, the system updates the user's proficiency vector to U_p^{t+1} by assessing their performance, provides actionable guidance tailored to their needs, and adapts future scheduling dynamically. The overarching objective of the system is to maximize learning gains and user engagement throughout their skill development journey. To achieve this objective, this system should address the challenges of scalable knowledge discovery, personalized practice scheduling, and pedagogical knowledge delivery.

4 The SocialCoach Framework

To address these challenges, we introduce **SocialCoach**, an agentic tutoring and practice system designed for personalized social skill training at scale. As illustrated in Figure 2, SocialCoach holistically integrates four critical components. Firstly, it automatically constructs a comprehensive and pedagogically-grounded social skills knowledge corpus from diverse, unstructured sources using LLMs. Second, it dynamically maintains a user's social profile, capturing each learner's attributes, behaviors, and evolving proficiency. Additionally, SocialCoach employs an adaptive practice scheduling policy trained with reinforcement learning to prepare scenario practices tailored to individual needs. For each practice, it delivers pedagogically grounded coaching through immersive practice, fine-grained assessment, and personalized feedback. We elaborate on each of these components below.

4.1 Automated Knowledge Discovery

Effective social skill training depends on the availability of high-quality, context-rich pedagogical content that reflects real-world interpersonal challenges. However, expert knowledge in this domain is often tacit and dispersed across unstructured sources such as books, articles, and media [11]. Manual curation of such knowledge is labor-intensive and difficult to scale, while direct LLM-based content generation risks unverifiability and hallucination [39]. These methods compromise knowledge scalability or pedagogical reliability. To overcome these challenges, we introduce an end-to-end, LLM-driven pipeline that automatically constructs a structured and trustworthy social knowledge corpus $\mathcal{K}$. This method ensures pedagogical traceability and supports diverse social skills at scale.

Figure 3: The proposed theory-to-practice social knowledge framework with multi-faceted categorization perspectives.

4.1.1 ***Theory-to-Practice Knowledge Organization.*** Effective social skills education requires more than isolated facts or generic tips; it demands a comprehensive framework that connects abstract principles, concrete examples, and hands-on practice. Most existing approaches treat these components in isolation, limiting transferability and depth of learning. We address this gap by designing a theory-to-practice knowledge framework across three pedagogical levels [37]. This method ensures learners not only understand what to do, but also why and how, leading to transferable social competence. This integration comprises three interrelated layers:

- *Strategic Theories* $\mathcal{K}_t$: Provides the conceptual "why" through abstract principles defining effective social interaction.
- *Illustrative Case* $\mathcal{K}_c$: Bridges theory and practice by showing the "how" through concrete examples and case studies that demonstrate social principles in action.
- *Practical Scenarios* $\mathcal{K}_s$: Enables the "doing" by immersing learners in hands-on simulations for active practice and assessment, thereby closing the "knowing-doing" gap.

See our GitHub resources for these data examples.

4.1.2 ***Multi-faceted Semantic Categorization.*** Social interactions are highly diverse and context-dependent, requiring not only an understanding of specific skills but also their flexible application across various scenarios. Thus, we introduce a multi-faceted semantic categorization method to structure the knowledge corpus. This method represents knowledge in a granular yet holistic way, facilitating flexible indexing and fine-grained retrieval of content based on each learner's unique needs. Specifically, each knowledge item is categorized along the following dimensions:

- *Social Competency* (C_c): Foundational social-emotional abilities based on widely-used CASEL theory [10], encompassing self-awareness, self-management, social awareness, relationship skills, and responsible decision-making.
- *Social Skills* (C_s): Actionable, specific skills such as conflict resolution, empathy, and negotiation. Drawing from the CASEL framework, we identify 34 essential social skills that support the five core competencies [1].
- *Scenario Context* (C^t): Contextual categories that are derived from social environments and relationship dynamics, such as workplace, romantic and public situations [18].

See Appendix A.1 for detailed descriptions of these categories.

4.1.3 ***LLM-Powered Corpus Construction Pipeline.*** To scalably extract pedagogical and traceable content of social knowledge, we introduce an automated pipeline leveraging LLM-powered multi-agent systems for efficient knowledge extraction and annotation [26]. This pipeline yields a structured and pedagogically grounded knowledge base, with the following principal key stages.

(a) *Curated Data Collection.* To ensure both pedagogical value and rigor, we strategically aggregate high-quality, diverse materials on social skill. These sources include authoritative books, academic papers, news articles, and other reputable resources, thereby ensuring coverage of established theories and authentic scenarios.

(b) *Schema-Guided Extraction.* We first define the schema to represent three types of social knowledge items (see Appendix A.1 for detailed schema description). Guided by this schema, an extraction agent systematically transforms the unstructured collected data into a set of structured knowledge entities.

(c) *Output Validation.* Then, we use a filtering agent to verify that the the extracted content adheres to the schema's format guidelines and the completeness across all required fields.

(d) *Multi-faceted Tagging.* To structure the knowledge semantically, a tagging agent annotates each knowledge item with multi-faceted categories, enabling nuanced organization.

(e) *Knowledge Refinement.* After tagging, a refinement agent reviews and revises both the content and metadata of each item to improve quality and accuracy. This step resolves inconsistencies, such as incomplete descriptions or incorrect tag assignments.

Through this multi-stage process, we construct a structured social knowledge corpus $\mathcal{K} = (\mathcal{K}^t, \mathcal{K}_c, \mathcal{K}_s)$, where each $k \in \mathcal{K}$ is associated with semantic labels $c_c \in C_c$, $c_s \in C_s$, and $c^t \in C^t$.

4.2 User Social Profiling

Personalized social skills training necessitates a comprehensive and adaptive understanding of each learner [31]. To this end, we construct a dynamic user profile U^t, which is continuously updated at each timestep t throughout the learning trajectory. This profile systematically encapsulates several key aspects, detailed as follows.

- *Personal Information* U_i^t. Each user is initially characterized by a set of attributes, including demographics (e.g., age, gender), personality (e.g., habits, interests), and targeted social skills $S_U \subseteq S$. These attributes are optionally self-reported during the onboarding process and are leveraged to tailor scenario practices and guidance content.
- *Historical Behaviors* U_b^t. The system continuously records a rich behavioral trace of each learner's interactions, including completed scenarios, reflection responses, engagement metrics (e.g., session frequency, time spent), and other relevant activity logs. These data enables the inference of learning patterns and provides contextual signals to inform both adaptive scheduling and formative assessment.

[1]https://casel.org/fundamentals-of-sel/what-is-the-casel-framework

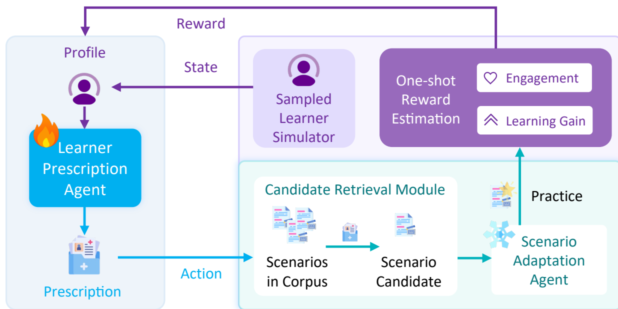

Figure 4: Scenario reference retriever optimization with reinforcement learning from learner simulation feedback.

- *Proficiency Status* U_p^t. To monitor learning progress, we maintain a dynamic vector representing the learner's proficiency on both high-level CASEL competencies and granular targeted skills. Each item is scored on a [1-5] scale, mapping to standard rubrics (i.e., 1=Novice and 5=Expert).

By integrating these dimensions, we build the dynamic user profile $U^t = (U_i^t, U_b^t, U_p^t)$ to support a personalized learning experience.

4.3 Adaptive Practice Scheduling via RL-Optimized Prescription Agent

The development of social skills is experiential, and effective progress hinges on presenting each learner with the right practice at the right time. While static curricula lack enough personalization, LLM-generated content risks pedagogical value. Inspired by the expert judgment of human coaches, we introduce an adaptive practice scheduling module as an instructional planner and further optimize it using agentic RL [41] to address the cold-start issue.

4.3.1 ***Prescription-Retrieval-then-Adaptation***. At each session t, we schedule a scenario practice p^t with the following stages.

Practice Prescription. Emulating expert pedagogical reasoning, a practice prescription agent analyzes the learner's current profile U^t to prescribe a scenario requirement to best support the learner's growth. This requirement consists of a brief query and optional scenario attributes used as retrieval filters (e.g., associated competencies, skills, types of contextual context, etc).

Candidate Retrieval. The prescribed requirement serves as a query and filters to retrieve the most suitable scenario from the social knowledge corpus K_s. We use a dual-retrieval combining tag-based filtering for pedagogical alignment with semantic search to identify the ideal candidate. If strict filters yield no results, we progressively relax to ensure a suitable candidate is retrieved.

Context Adaptation. Finally, a context adaptation agent personalizes the retrieved scenario. Based on the learner's profile U^t, the agent selects the most suitable role in the scenario for the learner's role-playing practice. This step maximizes the relevance and engagement while grounded in the trustworthy corpus.

See Appendix A.2 for the prescription schema and retrievability.

4.3.2 ***Representative Learner Simulation Environment***. To optimize for long-term, judge-rated pathway quality and user engagement, the prescription policy π necessitates high-level cognition of both learner's status and our scenario corpus, requiring training for adaptation. Lacking diagnosis misaligns prescriptions with real skill gaps, and lacking corpus awareness yields suboptimal retrieval of pedagogically aligned scenarios. However, such a real-world dataset is vacant, and collecting one is expensive and difficult to scale, which incurs the "cold-start problem". To mitigate this issue, we develop a simulation environment grounded in real-world data for efficient training, using role-playing capabilities of LLMs [4, 42, 49].

Representative Learner Profile Construction. To ensure the realism of simulated users, we sourced user data from two large-scale public datasets driven from real-world: SocioVorse [47] which includes million-level demographic data, and PersonaHub [13] which contains billion-level personality descriptions. We embed these items using Qwen3-Embedding-8B [48] to create semantic representations. We then apply the Maximal Marginal Relevance algorithm [3] to sample the top-n representative items from each set. For a highly efficient and deterministic matching process, we compute an $n \times n$ compatibility matrix between sampled demographic profiles and personality descriptions, where each entry is the dot-product similarity of their embeddings. We then apply the Hungarian algorithm [23] to obtain a one-to-one matching that assigns each demographic profile to exactly one personality description, maximizing total compatibility. The resulting matched pairs form representative, coherent synthetic learner profiles. Finally, for each of the n profiles, an LLM initializes 3–5 target skills, starting proficiency scores, forming a diverse and realistic simulated user base. Additionally, it also infers a learning pace α_U on a 1-5 scale (1 = slowest, 5 = fastest) and then normalizes to [0, 0.1], which modulates future proficiency score updates.

One-Shot Reward Estimation. Obviously, running a full multi-turn dialogue simulation for every training step is computationally expensive. Since advanced LLMs like GPT-5 [28] and Qwen3 [43] have demonstrated the ability to accurately track human judgment and simulate realistic social behaviors [16, 19, 45], we adopt a one-shot LLM-based proxy reward to facilitate efficient training and address the cold-start challenge. To ensure pedagogical reliability and prevent reward hacking, we implement an output validity bottleneck that assigns a zero score ($r^t = 0$) for any non-compliant or meaningless generation. For valid outputs, the reward is decomposed into two standardized rubrics: a learner-simulated Engagement score (r^t_{engage}) and a coach-simulated Learning Gain score (r^t_{gain}), both rated on a 1–5 scale. The final reward is formulated as $r^t = r^t_{\text{engage}} + r^t_{\text{gain}} + \eta$, where η represents a penalty factor specifically for empty retrieval results caused by overly restrictive filters. To further ensure training robustness and minimize model-specific idiosyncrasies, we employ GPT-5 with a temperature of 0 and utilize rubric-grounded checklists to standardize the scoring criteria.

4.3.3 ***Sequential Decision Formulation***. We formulate the sequential practice scheduling task as a Markov Decision Process (MDP) with the following key components:

- *State* (s^t): The learner's dynamic profile U^t, including personal information, behavior history and proficiency status.
- *Action* (a^t): The practice requirement prescribed by π.
- *Reward* (r^t): The reward from the simulation environment.

- *Transition* (P): The evolution of the proficiency vector. Specifically, the scores for the associated granular skills and cross-skill competencies of the practiced scenario are updated via $U_{p,k_s}^{t+1} = \min(5, U_{p,k_s}^{t} + \alpha_U \cdot r_{\text{gain}})$. Here, $\alpha_U \in [0, 0.1]$ is a normalized learning pace derived from the user's profile.
- *Discount Factor* γ balancing immediate and future reward.

The objective is to find an optimal prescription policy to maximize the expected cumulative reward, $E[\sum_{t=0}^{T} \gamma^t r^t]$ for each learner.

4.3.4 ***Policy Optimization from Simulation Feedback.***
Due to the scarcity of real-world human interaction, it is a popular way to leverage LLM-based simulated feedback as rewards to optimize the performance of the agent for human adaptation [32, 42, 45, 50]. The scenario prescription policy is parameterized as an LLM. During training, for a sampled learner profile, the policy observes the state s^t, and generates a prescription a^t. This action is used to create a practice scenario p^t, which is evaluated by the user simulator to produce a reward r^t and the next state s^{t+1}. These interaction trajectories (s_t, a_t, r_t, s^{t+1}) are collected into an experience memory. We then use the multiturn agentic RL [41] to fine-tune the LLM-based policy, which effectively learns to make pedagogically sound scheduling decisions.

4.4 Pedagogical Coaching Scaffolding

Effective social skill acquisition requires more than passive exposure to content. It demands an iterative loop of guided practice, diagnostic assessment, and personalized feedback grounded in pedagogical principles [1]. To address this, we introduce a three-stage coaching framework in SocialCoach that integrates immersive practice, proficiency assessment, and reflective tutoring. By linking theory to practice, SocialCoach is designed to support deeper comprehension and skill transfer, helping address the "knowing–doing" gap.

4.4.1 ***Immersive Scenario-Based Practice.***
At each learning session t, for the scheduled scenario k_s^t, the social skill learning begins with immersive, goal-driven simulations designed to replicate real-world challenges. These tailored scenarios ensure engagement by creating a gamified practice environment.

Goal-Driven Scenario Simulation. The practice sessions are structured as goal-driven interaction games, where learners assume specific roles within challenging contexts [51]. Each session tasks the learner with achieving defined social objectives, such as mediating a conflict or negotiating a compromise, while interacting with LLM-driven agents representing other participants. These agents dynamically adapt their behavior and dialogue to the learner's actions, ensuring a realistic and responsive simulation. Sessions conclude upon goal achievement or reaching the interaction limit, with conversation logs recorded as $L^t \leftarrow \text{simulation}(U^t, k_s^t)$.

Role-Playing Consistency Enhancement To maintain realism, we infer contextual features such as power dynamics, emotional tone, and implicit social norms of virtual agents based on extracted dialogues in the corpus. Then, we embed them into prompts to enhance the response consistency, aligning with their character roles and the scenario context, and enhancing the interaction immersion.

4.4.2 ***Attribution-Based Proficiency Assessment.***
Upon completing each practice session, SocialCoach conducts a granular proficiency assessment to identify the strengths, weaknesses, and underlying deficits. This diagnosis combines explicit behavior analysis with reasoning-driven attribution to generate actionable insights.

Social Behavior Diagnosis. Using an LLM-powered diagnostic agent, SocialCoach analyzes conversation logs L^t to identify explicit strategies (e.g., active listening) and implicit reasoning (e.g., emotional awareness), i.e., $D_b^t = \text{LLM}_{\text{diagnosis}}(U^t, L^t)$. The agent detects and categorizes both positive and negative behaviors demonstrated during the simulation. These behaviors are mapped to CASEL competencies and targeted social skills, which enables precise identification of skill gaps and areas of improvement.

Reasoning-Driven Performance Attribution. Beyond surface-level assessment, SocialCoach introduces a diagnostic attribution mechanism to uncover the root causes behind observed negative behaviors, i.e., $D_c^t = \text{LLM}_{\text{attribution}}(U^t, L^t, D_b^t)$. Specifically, the LLM-based agent analyzes why a learner exhibited certain actions, categorizing these deficits into two types: *acquisition deficits* (insufficient understanding of relevant social concepts or skills due to lack of knowledge) and *performance deficits* (difficulty applying known skills effectively in the contexts despite possessing the necessary knowledge), grounded in social behavior disorders theory [14]. By disentangling these underlying causes, this analysis addresses not just "what went wrong" but "why it went wrong." Then, we update the learner's proficiency vector to reflect their evolving skill profile, i.e., $U_p^{t+1} = \text{LLM}_{\text{profiling}}(U^t, L^t, D_b^t, D_c^t)$.

4.4.3 ***Knowledge-Grounded Reflective Tutoring.***
To close the coaching loop, SocialCoach delivers reflective, actionable feedback tailored to the learner's specific needs. This stage aims to foster learners' deep understanding and transferable skill application.

Targeted Knowledge Retrieval. Based on the diagnostic assessment, SocialCoach retrieves relevant materials based on both tag-based filter and semantic embedding from the structured knowledge corpus, $\mathcal{K}^t = (\mathcal{K}_t^t, \mathcal{K}_c^t) \subset \mathcal{K}$. For identified *performance deficits*, illustrative cases $\mathcal{K}_c^t \subseteq \mathcal{K}_c$ are provided to demonstrate skill application in analogous real-world contexts. Additionally, for *acquisition deficits*, foundational theories $\mathcal{K}_t^t \subseteq \mathcal{K}^t$ are also retrieved to strengthen conceptual understanding.

Reflective Guidance Generation. Rather than just static feedback D^t and $\mathcal{K}^t$, SocialCoach further adopts Socratic questioning [25] to foster critical reflection, i.e., $G^t = \text{LLM}_{\text{guidance}}(U^t, L^t, D_b^t, D_c^t, \mathcal{K}^t)$. For example, it might ask: "Back to the situation of a case, what alternative strategies could you have employed to de-escalate the conflict?" This approach enables a reflective exercise, encouraging learners to analyze their actions and explore alternative strategies.

5 Experiments

We conduct a multi-part empirical study of SocialCoach to assess the quality of its knowledge corpus, the judge-rated quality of its adaptive scenario scheduling, and the perceived value of its pedagogically grounded coaching. Due to the page limit, please see more resources like code, data examples, agent prompts, output examples, and so on in Github: https://github.com/GeminiLight/SocialCoach.

Table 1: Corpus Statistics and Tag Diversity

Type	Overall Statistics		Semantic Tag Diversity		
	Size	Avg. Words	Competence	Skill	Context
Theory	14,186	150.21	52.84	4.55	-
Case	23,521	86.92	54.78	4.69	2.43
Scenario	5,463	372.45	43.51	4.45	2.66

Table 2: Expert Likert Ratings

Type	Tag Annotation Agreement			Content Pedagogical Value		
	Competence	Skill	Context	Soundness	Accessibility	Pedagogy
Theory	4.78	4.62	-	4.08	3.81	3.26
Case	4.68	4.55	3.86	4.06	3.98	3.40
Scenario	4.92	4.82	4.63	4.83	4.41	4.44

5.1 System Implementation

5.1.1 ***Data Collection Method.*** For the research goal in this study, we construct the social knowledge corpus by curating books from recognized lists and category rankings. First, we identify relevant books using category rankings from platforms like Amazon Books' self-help section[2] (e.g., relationships, emotional intelligence) and BookAuthority.org[3] (e.g., leadership, empathy, social intelligence). Then, we gather the available files of these books from the Open Library website[4]. After a manual review for relevance and quality, we selected 200 books to ensure a focused and comprehensive corpus. The final corpus includes many widely cited works such as "Nonviolent Communication: A Language of Life" and "Emotional Intelligence: Why It Can Matter More Than IQ".

5.1.2 ***System Implementation Details.*** We support the various LLMs as the backend of agents, such as GPT-5 [28], Qwen3-235B [43], DeepSeek-R1-0528 [8], Gemini-3-Pro, Qwen3-8B, etc. Concretely, for the corpus construction pipeline, we employ Qwen3-235B as the backend, Chroma as the vector database, and Qwen3-Embedding-8B [48] as the embedding model. Next, regarding the training of a prescription agent, we adopt the Qwen3-8B as a base LLM and train it with multi-turn agentic RL [36] supported by ROLL [41]. These training experiments are conducted on a cluster of 16 NVIDIA H20 GPUs with a learning rate of 1.0×10^{-6} and a maximum output length of 2048 tokens. We configure the parallel environment manager with a batch size of 64. In simulation environments, we set the number of representative profiles n to 1000 to ensure profile variety and max turns to 10. For reward shaping, we use Qwen3-235B with a temperature of 0 for one-shot estimation and consider the filter penalty η as 0.1. Regarding the candidate retrieval module, it combines boolean filtering for pedagogical compliance with semantic ranking via Qwen3-Embedding-8B [48]. If filters yield no results, a progressive relaxation strategy systematically eases constraints to ensure a candidate is always surfaced. Finally, for coaching scaffolding, this service layer supports multiple LLMs and uses GPT-5 by default.

[2]https://www.amazon.com/amz-books/store
[3]https://bookauthority.org/categories
[4]https://openlibrary.org

5.2 Analysis of Knowledge Corpus

Our automated pipeline produces a corpus of over 40,000 unique knowledge entries, each tagged with multifaceted semantics. To assess the quality, diversity, and pedagogical value of this corpus, we conducted the following multi-faceted analysis.

5.2.1 ***Overall Corpus Statistics.*** As shown in Table 1, this corpus is composed of 14,186 theories, 23,521 cases, and 5,463 scenarios. We analyze the textual depth of each category by calculating the average word count, revealing that scenarios are considerably longer than theories and cases. This difference reflects their pedagogical roles: scenarios provide detailed immersive practice, while theories and cases offer concise conceptual knowledge and examples.

5.2.2 ***Tag Diversity & Distribution.*** To estimate whether the corpus is well-balanced, we evaluated the diversity of our semantic tags using Shannon Entropy, which measures the uniformity of a distribution. As shown in Table 1, the high entropy scores of each knowledge type suggest that the corpus covers a broad range of social situations rather than being limited to a few contexts. A detailed breakdown of the tag distributions is as follows:

5.2.3 ***Expert Likert Ratings.*** To assess the tag accuracy and pedagogical value, three expert reviewers who are workplace practitioners proficient in social skills and emotional intelligence rated a random sample of 150 corpus entries (50 for each type of knowledge) on a 5-point scale from multiple dimensions. The manual validation yielded an average Fleiss' Kappa of 0.63, indicating moderate-to-substantial agreement across assessments. Results suggest high annotation consistency and pedagogical value within the sampled entries. See Appendix B.2 for detailed analysis.

5.3 Evaluation on System Quality

To assess the judge-rated quality of SocialCoach's core components, we conduct experiments within our learner simulation environment. We first generate a held-out set of 200 synthetic learner profiles for testing, ensuring they are distinct from any profiles used during the RL training. Each profile is initialized with a unique combination of demographics, personality traits, target social skills, and initial proficiency levels to represent a diverse user population. Following existing studies [16, 40, 45], we adopt GPT-5 [28] as the judge on the 5-point Likert scale assessment to evaluate output quality.

5.3.1 ***Visualization of Profile Distribution.*** We visualize the distribution of representative profile embedding to show their consistency and diversity. See Appendix B.3 for details.

5.3.2 ***Evaluation on Practice Scheduling.*** To evaluate our adaptive practice scheduling, we simulate 10-session learning pathways for each simulated learner. We compare our RL-finetuned SocialCoach agent against baselines, including a ProfileQuery that regards the user profile as a query for a semantic retrieval method and several NonTuning LLMs that directly generate a query without finetuning. We assess the pathway quality across four criteria: Avg. Engagement (relevance to user goals), Avg. Learning Gain (pedagogical value), Personalization (tailoring to user background), and Progression (coherence of the learning sequence on progress).

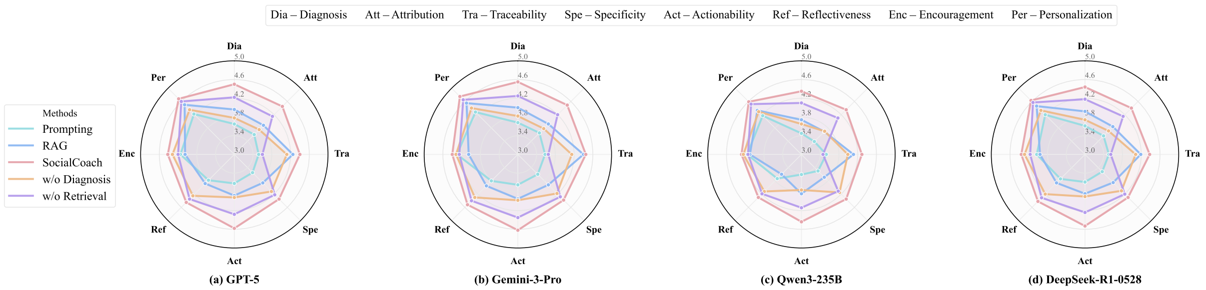

Figure 5: Evaluation results on tutoring guidance.

Table 3: Evaluation results on practice pathway.

Method	Eng.[1]	LG[2]	Pers.[3]	Prog.[4]
ProfileQuery	3.45	3.52	3.61	3.40
NonTuning$_{\text{GPT-5}}$	3.88	3.75	4.02	3.74
NonTuning$_{\text{Gemini-3-Pro}}$	3.95	3.88	4.08	3.78
NonTuning$_{\text{Qwen3-235B}}$	3.92	3.68	3.96	3.67
NonTuning$_{\text{DeepSeek-R1-0528}}$	3.76	3.72	3.92	3.68
NonTuning$_{\text{Qwen3-8B}}$	3.61	3.44	3.78	3.42
SocialCoach$_{\text{Qwen3-8B}}$	**4.38**	**4.11**	**4.25**	**4.05**

[1]Avg. Engagement; [2]Avg. Learning Gain; [3]Personalization; [4]Progression.

As shown in Table 3, SocialCoach$_{\text{Qwen3-8B}}$ receives higher judge ratings across all four evaluation criteria, particularly on Engagement and Learning Gain. While powerful zero-shot models like NonTuning$_{\text{GPT-5}}$ and NonTuning$_{\text{Gemini-3-Pro}}$ perform reasonably well, the observed gap between them and the finetuned SocialCoach suggests a limitation of non-adapted approaches. Even if a strong LLM generates a pedagogically sound prescription, it may fail to retrieve a highly rated practice if that prescription is not well-aligned with the available scenarios in the knowledge corpus. This suggests that practice quality relies not only on an accurate understanding of the learner's status but also on generating prescriptions that are realistically retrievable. This indicates that our training adaptation may improve the agent's awareness of both the user's diagnosed needs and the specific contents of the corpus, resulting in better-aligned prescriptions for practice retrieval.

5.3.3 ***Evaluation on Pedagogical Tutoring***. To evaluate the pedagogical feedback, we use GPT-5 [28] to simulate a learner's behavior across 200 practice sessions. We compare SocialCoach against four baselines: (a) Prompting LLM that provides feedback directly, (b) RAG model that retrieves from our corpus, and two ablated variants of our system: (c) one without assessment workflow, and (d) another without retrieval module. We adopt eight metrics from detail to whole to comprehensively evaluate the guidance quality: Diagnosis (accuracy of problem identification), Attribution (understanding of root causes), Traceability (faithfulness to the knowledge source), Specificity (relevance to user actions), Actionability (clarity of suggested improvements), Reflectiveness (use of Socratic questioning), Encouragement (supportive and positive tone), and Personalization (tailoring to the user's profile).

As shown in Figure 5, the results show that SocialCoach receives higher scores than the baselines and ablated models across most metrics. Regarding the Diagnosis and Attribution, SocialCoach receives higher ratings on these two metrics compared to baseline and w/o Diagnosis variant. This suggests an important role of the attribution-based assessment workflow. This module is designed to help the system move beyond surface-level feedback by identifying skill gaps and their possible root causes in a fine-grained manner. Additionally, we observe that both retrieval-based methods, such as SocialCoach and RAG, achieve high scores in Traceability. This suggests that anchoring feedback in a verifiable knowledge corpus can improve perceived reliability. Furthermore, SocialCoach also receives higher scores on more integrative metrics, such as Actionability and Personalization. This supports the value of combining component design with educational domain knowledge. Overall, these results suggest that SocialCoach can deliver personalized, pedagogical guidance within a comprehensive coaching loop under our simulated and judge-rated evaluation setting.

5.3.4 ***Alignment with Human Evaluation***. To estimate the reliability of our LLM-based evaluations, we compare them against independent human judgments across key metrics. The alignment of most metrics indicates that the automated assessments are moderately correlated with expert ratings, while some subjective metrics remain challenging. See Appendix B.4 for details.

5.3.5 ***Case Study of Learning Journey***. In addition to our quantitative analysis and LLM-based evaluations, we further provide case studies in Appendix B.5, which trace a concrete learning journey to illustrate system behavior and perceived pedagogical value.

5.4 End-to-end User Study

5.4.1 Practical Product Deployment. We have deployed SocialCoach in our product, EQoach. This application features a mobile-first React frontend and a Python backend. SocialCoach serves as its core engine via algorithmic APIs that power adaptive practice scheduling, the LLM-driven simulation arena, and the coaching loop. We provide its key interfaces in Appendix C.1. An internal beta of EQoach launched in April 2025, which offers beta testers a gamified environment for social skill learning.

5.4.2 User Study Analysis. Given the emerging nature of our EQoach product, its user scale is still limited. To collect early user-experience and perceived-usefulness feedback, we opt for a 10-practice user study under strict privacy constraints. We recruit 50 participants (25 university students, 25 early-career professionals; aged 20–30), representing key demographics for social skill development. Participants were instructed to use EQoach for at least 10 practices.

At the end of the study period, we administered a post-study questionnaire and conducted semi-structured interviews with them to explore their experiences. See Appendix C.2 and C.3 for analysis and simulator fidelity validation.

6 Conclusion

In this paper, we introduced SocialCoach, an LLM-powered agentic tutoring and practice framework that supports personalized social skill learning. We first constructed a pedagogically grounded, theory-to-practice social knowledge corpus at scale. Then, we optimized adaptive, goal-aligned scenario scheduling via a prescription–retrieval–adaptation policy trained with simulation-based RL to personalize practice. Furthermore, we integrate immersive simulations with attribution-based proficiency assessment and knowledge-grounded reflective tutoring to address the knowing–doing gap. Finally, we deployed it in EQoach and conducted corpus analyses, simulated evaluations, judge-rated tutoring evaluations, and an early user study. The results suggest promising knowledge quality, pathway personalization, judge-rated engagement, perceived user value, and practical viability. Overall, as a holistic agentic system, SocialCoach provides a practical architecture for personalized and gamified pedagogical platforms for soft skills learning.

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A Details of Framework Design

A.1 Knowledge Categorization Details

To ensure our knowledge corpus is pedagogically sound and easily retrievable, we structure it using a multifaceted categorization method. Each piece of knowledge is tagged along three key dimensions: its associated CASEL competency, the specific social skill it addresses, and the scenario type in which it applies.

A.1.1 ***Associated Social Competencies.*** In this work, we adopt the Collaborative for Academic, Social, and Emotional Learning (CASEL) framework [10], which is recognized as one of the most popular SEL frameworks by the Explore SEL project[5] of Harvard Graduate School of Education. CASEL provides a widely acknowledged foundation for understanding social-emotional learning and identifies five core competencies: Self-Awareness, Self-Management, Social Awareness, Relationship Skills, and Responsible Decision-Making. These competencies represent the broad, overarching abilities crucial for navigating social interactions effectively.

A.1.2 ***Associated Social Skills.*** To bridge the gap between CASEL's abstract competencies and practical application, the CASEL office further categorizes knowledge by associated Social Skills. These are defined as granular, observable, and teachable behaviors—such as specific communication techniques or actions—that individuals can practice to demonstrate proficiency within the broader CASEL domains. Our system tags content with these fine-grained skills to

[5] http://exploresel.gse.harvard.edu/compare-domains/

Listing 1: An example of a structured prescription.

```
{
  "prescription": {
    "query": "Navigating a technical disagreement with a
         senior researcher during a project meeting to
         find a compromise.",
    "filters": {
      "associated_competencies": {
        "values": ["Relationship Skills", "Responsible
             Decision-Making"],
        "logic": "OR"
      },
      "social_skills": {
        "values": ["Conflict Resolution", "Teamwork"],
        "logic": "AND"
      },
      "situational_contexts": {
        "values": ["Workplace", "Office"],
        "logic": "OR"
      }
    }
  }
}
```

enable the precise targeting of learning objectives and assessment criteria. The relationship between CASEL competencies and their corresponding social skills is detailed in Table 4.

A.1.3 ***Scenario Contextual Types.*** Since the usefulness of a social skill often depends on its context, we also categorize scenarios by Scenario Type. This categorization is hierarchical to provide nuanced classification. First, we assign a high-level category describing the general environment where the interaction occurs (e.g., Workplace, Family). Next, we refine this with a more detailed sub-category (e.g., Office, Parent-child). This hierarchical approach, detailed in Table 5, helps learners practice skills in contexts directly relevant to their personal and professional lives, with the goal of supporting transfer across contexts.

A.2 Prescription Schema and Retrievability

We provide a formal description of the prescription action schema and the retrievability mechanism for scenario practices.

A.2.1 Action Schema Components. The action a^t generated by the prescription agent is formalized as a structured scenario requirement designed to emulate the pedagogical reasoning of a human coach. This requirement comprises a natural language query describing the core learning objective alongside a filter set that leverages boolean logic to navigate the multi-faceted semantic organization. Specifically, the system maintains a default AND operation across the distinct dimensions of associated competencies, social skills, and situational contexts to ensure that retrieved practices satisfy all necessary pedagogical and environmental constraints simultaneously. Within specific dimensions, the agent's output schema supports both AND and OR operations to allow for either precise targeting or flexible grouping of related concepts. For example, as illustrated in Listing 1, the agent may require a strict AND match for "Conflict Resolution" and "Teamwork" to ensure both skills are practiced, while allowing an OR match for "Workplace" or "Office" contexts to maintain high retrievability.

Table 4: The list of social skills grouped by their main related competencies under the CASEL framework.

Competencies	Social Skills
Self-Awareness	Identifying emotions
	Social and cultural identity
	Recognizing strengths
	Growth mindset
	Self-efficacy
	Examining bias
	Sense of purpose
Self-Management	Emotion regulation
	Impulse control
	Stress management
	Self-discipline and motivation
	Perseverance
	Goal-setting
	Organizational skills
	Initiative and Agency
Social Awareness	Perspective-taking
	Empathy and compassion
	Expressing gratitude
	Appreciating diversity
	Identifying social norms and demands
	Sense of belonging
Relationship Skills	Communication
	Cultural competence
	Building relationships
	Teamwork and working cooperatively
	Resolving conflicts
	Helping/Seeking help
	Leadership
	Standing up for the rights of others
Responsible Decision-Making	Demonstrating curiosity and open-mindedness
	Identifying and solving problems
	Analyzing situations and consequences
	Ethical responsibility
	Reflecting on one's role to promote individual and collective well-being

A.2.2 Retrievability Guarantee. Strict filtering criteria can lead to empty retrieval results, potentially disrupting the pedagogical flow. To ensure a robust retrievability guarantee, the candidate retrieval module employs a progressive relaxation strategy. If initial boolean filters yield no matches, the system systematically eases categorical constraints by transitioning from strict AND to OR logic within categories or progressively removing filter dimensions until a suitable candidate is identified. This ensures that even when

Table 5: 7 scenario contextual categories detailed in 26 types.

Contextual Category	Detailed Contextual Types
Workplace	Office, meetings, job interviews, professional collaboration
Family	Parent-child, siblings, home interactions, Spouse
Friendship	Informal, emotional bonding, casual talk
Romantic	Dating, partner conflicts, affection sharing
Education	Classroom, peer discussion, teacher-student roles
Public/Stranger	Store, transport, unknown people, Park, Courtroom
Party/Social	Events, birthdays, wine parties, social mingling

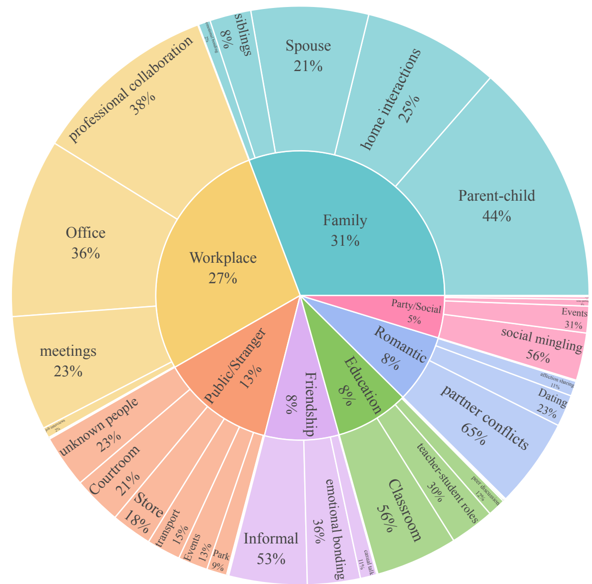

Figure 6: Distribution of scenario contextual types.

specific metadata matches are scarce, a traceable and pedagogically relevant practice candidate is always surfaced.

B Experimental Results and Analysis

B.1 Visualization on Corpus Distribution

Social Competency Co-occurrence. The heatmap in Figure 7 reveals logical correlations between competencies. We observe two distinct clusters with strong positive correlations (scores > 0.3): one linking Social Awareness with Relationship Skills, and another connecting Self-Awareness, Self-Management, and Responsible Decision-Making. This indicates that our corpus effectively captures the synergistic nature of social skills, where understanding others is tied to interaction, and personal insight is linked to self-regulation and thoughtful action.

Social Skill Frequency. Figure 8 displays the frequency of the most common social skills within our corpus. Foundational abilities such as communication, empathy and compassion, and Perspective-taking are the most prevalent.

Scenario Context Distribution. The distribution of scenario contexts, visualized in Figure 6, is broadly balanced across key life domains. The most prominent categories are Family (31%) and Workplace (27%), with coverage of other areas like public and educational settings. This balance suggests the corpus provides contextually relevant scenarios for a range of real-world interpersonal challenges.

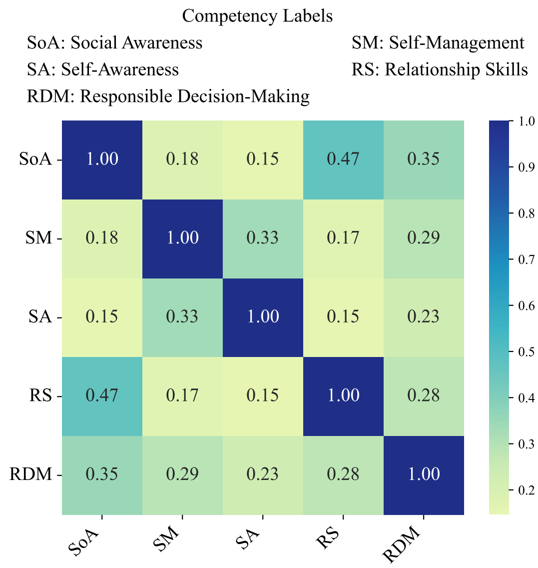

Figure 7: Co-occurrence heatmap of social competencies

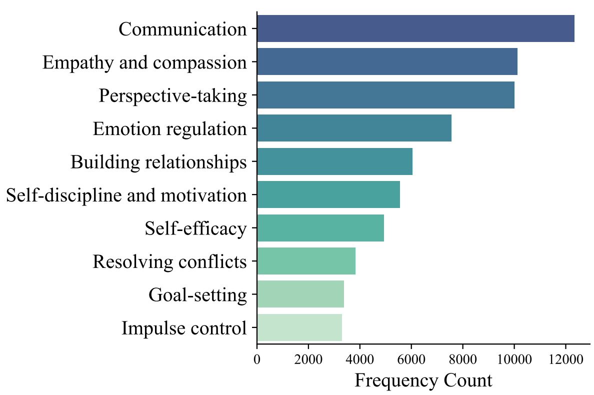

Figure 8: Top 10 social skills by frequency.

B.2 Expert Likert Ratings of Corpus

Tag Annotation Consistency. To assess the reliability of the automated tagging process, we measure the label agreement of human experts on these automated tags. As shown in Table 2, the expert evaluations show consistently high agreement scores across three dimensions. The agreement on scenarios achieves the highest ratings, which reflects the system's exceptional ability to tag structured, context-rich content accurately. Theories and cases, while foundational, scored slightly lower for context (3.98 and 3.86) due to their more abstract nature. These great results validate the reliability of the tagging system in categorizing actionable and relevant information, ensuring effective organization.

Content Pedagogical Value. To assess the content quality, these experts rate the knowledge content from the following aspects: Soundness (assesses the credibility and authenticity), Accessibility (assesses how clear the content is for a typical learner), and Pedagogy (assesses the potential usefulness for achieving a meaningful

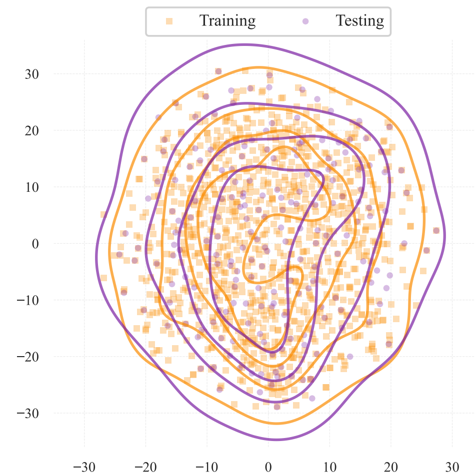

Figure 9: Representative learner profile distribution.

Table 6: Evaluation correlation between Human and LLM.

Category	Metric	Correlation	p-value
Learning Path	Learning Gain	0.51	$p < 0.01$
	Engagement	0.37	$p < 0.05$
	Personalization	0.48	$p < 0.01$
	Progression	0.42	$p < 0.05$
Tutoring Guidance	Diagnosis	0.48	$p < 0.01$
	Attribution	0.39	$p < 0.01$
	Traceability	0.52	$p < 0.01$
	Specificity	<u>0.21</u>	<u>$p > 0.05$</u>
	Actionability	0.45	$p < 0.01$
	Reflectiveness	0.36	$p < 0.05$
	Encouragement	0.34	$p < 0.05$
	Personalization	0.41	$p < 0.01$

learning outcome). As shown in Table 2, the results highlight the high pedagogical quality of the corpus across three dimensions. While scenario ratings consistently achieved the highest, the foundational theory scores are slightly lower. This indicates room for improvement in making abstract principles more engaging and instructionally impactful. Overall, our corpus delivers credible, clear, and effective content for social skill development, with scenarios standing out as particularly valuable for experiential learning.

B.3 Visualization on Profile Distribution

We visualize the distribution of representative learner profile embeddings using t-SNE, shown in Figure 9, The alignment between training and testing clusters suggests that the construction method effectively captures entire and consistent patterns across both datasets. Both clustering patterns highlight that both training and testing profiles reflect diverse learning needs and preferences, which is critical for real-world persona simulation.

B.4 Alignment with Human Evaluation

To validate our automatic evaluation, we measured the Pearson correlation between our LLM judge and ratings from three human

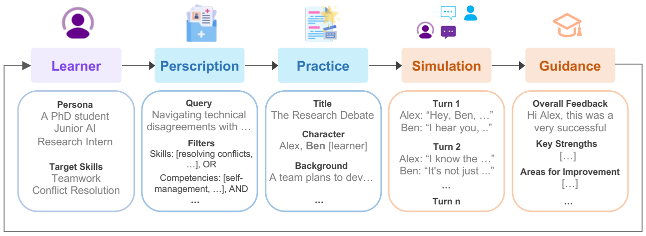

Figure 10: An illustrative learning journey case.

experts on 50 sample outputs. As shown in Table 6, the LLM judge shows moderate alignment with human ratings on several criteria. Significant positive correlations for metrics like Traceability, Learning Gain, Diagnosis, and Actionability suggest that it can serve as a useful proxy for assessing factual grounding, educational value, and analytical depth. Conversely, the alignment is weaker for subjective metrics. The weak correlations for Progression and other nuanced aspects suggest LLMs struggle to fully capture the subtleties of pedagogical flow and user experience, which often depend on subjective human judgment.

B.5 Case Study of Learning Journey

We illustrate a case study in Figure 10, featuring a learner whose persona is a junior AI research intern. This learner, while technically proficient, struggles with navigating professional disagreements and seeks to improve Teamwork and Conflict Resolution skills.

Based on this profile, the prescription agent generates a targeted query for a scenario about "navigating technical disagreements with peers," using filters to find content tagged with the Relationship Skills and Responsible Decision-Making competencies. The system retrieves a highly relevant scenario from the corpus titled "The Research Debate." In this practice, two researchers must resolve a conflict over choosing between a high-risk, innovative solution and a safer, established one under a tight deadline. Then, the adaptation agent assigns the learner the role of "Alex," the character advocating for the more ambitious, innovative approach.

In the simulation, the learner (as Alex) moves from defending his position to resolving the conflict. After validating his colleague's concerns about the project timeline, he proposes a concrete compromise: a "two-day spike" to test the new solution without risking the schedule. They reached a consensus.

Upon completion, the system provides a detailed guidance report. The feedback praises the learner's effective use of validation and integrative negotiation to reach a win-win outcome. It also identifies an area for growth: asking clarifying questions earlier to bypass the initial defensive exchange. Finally, the report poses reflective questions to help the learner consider how to establish shared goals. This process forms a complete learning loop.

C User Study via Deployed Product

C.1 Description of Product Interfaces

We deployed SocialCoach in our product, EQoach, and its key interfaces are presented in Figure 11. After onboarding for initial profiling, the user journey begins in the practice arena, where learners are presented with a variety of scenarios like "Managing Tensions in a Meeting". SocialCoach's adaptive module dynamically schedules these scenarios to match the user's specific learning needs. Upon selecting a practice, the learner is briefed on the scenario's background, the characters involved (e.g., "Sam" and "Anna"), and the social objectives they need to achieve. The learner then enters an immersive simulation, engaging in a goal-driven conversation with LLM-powered agents who respond dynamically to their actions. This interactive dialogue serves as the basis for the subsequent attribution-driven proficiency assessment and knowledge-grounded reflective tutoring. The gamified design of EQoach illustrates SocialCoach's ability to deliver an engaging learning experience that integrates adaptive practice with a complete pedagogical coaching loop.

C.2 Analysis of User Study

Results are shown in Figure 12; we summarize the main patterns below.

Adaptive and Targeted Scheduling. Participants highly valued the system's ability to align practice with individual needs. *Pedagogical Alignment* was rated at 4.66 ± 0.55, suggesting that users perceived the system as matching social competencies to personal growth goals. *Curriculum Fluidity* followed at 4.38 ± 0.82, with users noting that the adaptive pathways provided a logical progression through increasingly complex social challenges. One participant noted, "The clear guidance made it easier to navigate addressing complex tasks and achieve goals."

Immersive Practice and Realism. Participants perceived the simulation environment as realistic. *Agent Realism* received a high rating of 4.40 ± 0.66, highlighting the perceived consistency of the AI agents' personas and their natural responses. *Situational Fidelity* was rated at 4.32 ± 0.73, indicating that users felt the scenarios captured pressures similar to real-world social environments. As one participant stated, "The interactions felt surprisingly real; I felt the same level of social pressure as I would in a physical workplace conflict."

Diagnostic and Reflective Guidance. The reflective tutoring components achieved the highest marks in the study. *Reflective Depth* stood out with a rating of 4.88 ± 0.58, suggesting that users found Socratic questioning useful for analyzing their strategies. *Diagnostic Accuracy* also scored high at 4.46 ± 0.67, suggesting that users perceived the diagnostic feedback as meaningful. "The feedback helped me discover the 'why' behind my behavior," a participant observed.

Perceived Learning Value. Participants reported perceived growth in their social-emotional competence. *Skill Transferability* was rated at 4.56 ± 0.67, with users expressing confidence in applying practiced strategies to their professional lives. *Engagement Retention* also received a 4.48 ± 0.75, suggesting that the gamified loop helped maintain user motivation. One final interview revealed, "EQoach transformed my understanding of social intelligence into actionable skills I can use every day."

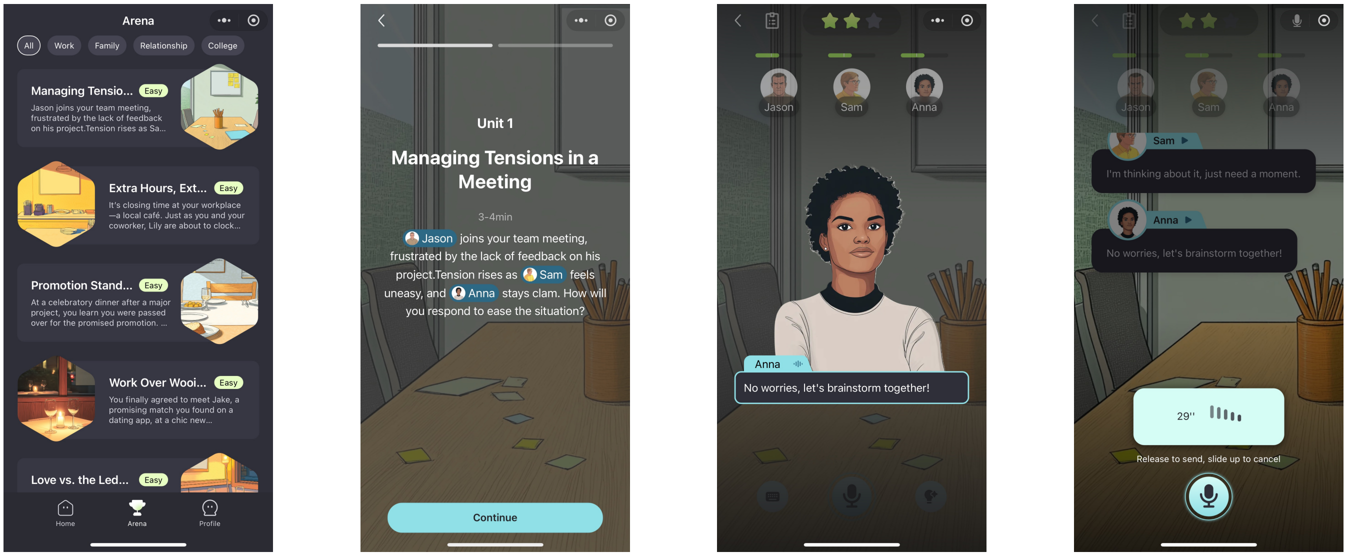

Figure 11: Several interfaces of our product, EQoach.

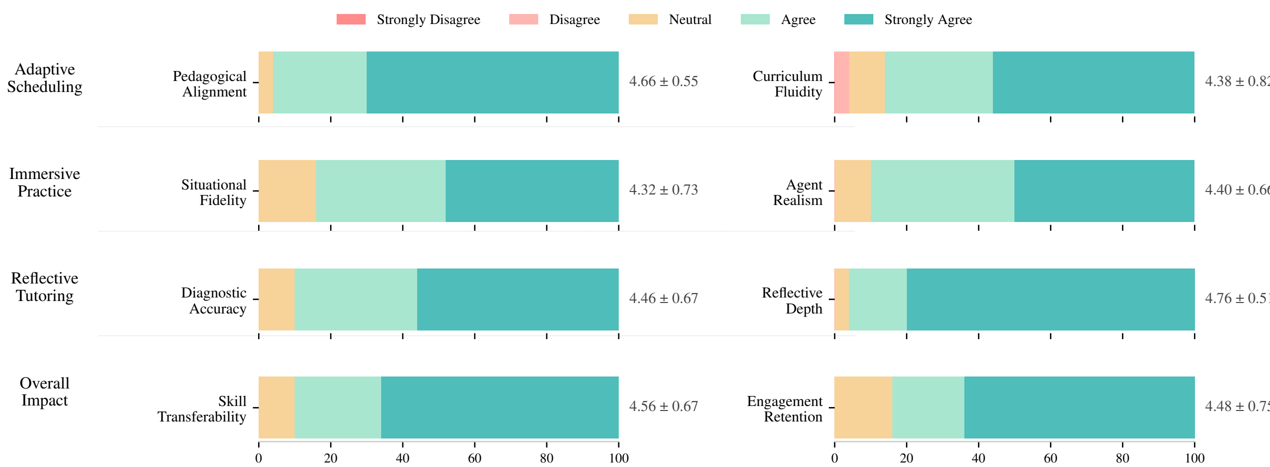

Figure 12: Questionnaire results from 50 participants.

C.3 Validation of Simulator Fidelity

To evaluate the reliability of our simulation environment, we conducted a correlation study comparing the one-shot reward estimation generated by GPT-5 [28] against independent human assessments. During the user study, learners provided the *Engagement* score from their own perspective, while human experts rated *Learning Gain* from a pedagogical coaching perspective, using the same 1–5 Likert scale as the simulator. We collected 100 samples, and the results revealed significant positive correlations between the simulator's rewards and expert judgments. Specifically, *Learning Gain* achieved a Pearson correlation of 0.42 ($p < 0.01$), while *Engagement* reached 0.37 ($p < 0.05$). These results suggest that the automated reward signal captures part of the underlying pedagogical value and user interest. This alignment indicates that one-shot estimation can serve as a useful but imperfect proxy for optimizing the RL policy toward meaningful educational outcomes.

D Further Discussion

D.1 Licensing and Legal Compliance

To address compliance and reproducibility concerns, we adopt a transformative, non-consumptive research framework for acedamic goal in this work. The knowledge corpus is constructed by extracting structural pedagogical patterns from 200 high-quality books accessed via the Open Library, a platform dedicated to digital lending for scholarly purposes. Rather than reproducing copyrighted expressive content, our pipeline transforms raw text into original, structured entities, i.e., theories (K_t), cases (K_c), and scenarios (K_s), that are functionally distinct from the source material. This process aligns with transformative fair use standards for text and data mining in academic research.